

Near-Orthogonality of Layer Jacobians in Deep Residual Networks

A Random-Matrix Proof with Explicit Constants

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Abstract

We prove that in an L -layer residual network satisfying a depth-scaled spectral-norm bound on residual block Jacobians, the per-layer contributions to the full parameter Jacobian are asymptotically *near-orthogonal* in the Frobenius inner-product sense. Concretely, for distinct layers $l \neq l'$,

$$|\langle J_\theta^{(l)}, J_\theta^{(l')} \rangle_F| \leq e^{2c} C_{\text{HW}} \sqrt{d_{\mathcal{I}} \cdot W} \quad \text{with probability } \geq 1 - 2e^{-d_{\mathcal{I}}},$$

while each diagonal term satisfies $\|J_\theta^{(l)}\|_F^2 = \Theta(e^{2c} d_{\mathcal{I}} W / L)$. The cross-to-diagonal ratio is therefore $O(L / \sqrt{d_{\mathcal{I}} W})$, which vanishes as $W \rightarrow \infty$ with L and $d_{\mathcal{I}}$ fixed. As a corollary, the effective rank of the Gauss–Newton matrix $G = J_\theta^\top J_\theta$ scales as $\Omega(L \cdot W_{\min} / d_{\mathcal{I}})$, growing multiplicatively with both depth and width. This closes the rank-summation gap present in prior treatments of the loss landscape of overparameterised residual networks and provides the first explicit constant for the cross-term bound. The proof combines the Hanson–Wright inequality [Rudelson & Vershynin, 2013] with a product-integral representation of the residual pushforward operator.

1 Introduction

A central empirical puzzle of deep learning is that gradient-based optimisers rarely become trapped at suboptimal local minima, even for highly non-convex loss functions. For overparameterised networks, several lines of work establish that the *existence* of bad local minima is structurally improbable, typically by showing that the Gauss–Newton matrix $G = J_\theta^\top J_\theta$ carries a high-rank positive-semidefinite component that dominates the residual curvature [Kawaguchi, 2016, Du et al., 2019, Oymak & Soltanolkotabi, 2020].

These treatments share a common gap. When the full parameter Jacobian J_θ is assembled by stacking per-layer contributions $\{J_\theta^{(l)}\}_{l=1}^L$, the effective rank of G is bounded not by the *sum* of per-layer ranks but by the rank of their union. The difference is decisive: if the per-layer Jacobians are highly correlated—sharing optimisation directions—depth provides

no advantage beyond parameter count. If they are near-orthogonal, ranks sum, and depth multiplies rather than merely adds to optimisation capacity.

This paper proves the near-orthogonality claim rigorously and provides explicit constants. The result is stated for deep residual networks satisfying a standard depth-scaling assumption on residual block Jacobians (Assumption 1), which is implied by gradient clipping or weight normalisation—both ubiquitous in practice.

Main result (informal). For any two layers $l \neq l'$ in an L -layer overparameterised residual network, the Frobenius inner product of their parameter Jacobians satisfies

$$|\langle J_{\theta}^{(l)}, J_{\theta}^{(l')} \rangle_F| = O(\sqrt{d_{\mathcal{I}} \cdot W} / L)$$

with high probability over initialisation, where $d_{\mathcal{I}}$ is the intrinsic dimension of the data manifold and W is the layer width. Each diagonal term is $\Theta(d_{\mathcal{I}}W/L)$, so the cross-term is o of the diagonal by a factor of $1/\sqrt{d_{\mathcal{I}}W}$. In the overparameterised regime $W \gg d_{\mathcal{I}}$, this ratio is small, confirming near-orthogonality.

Implications. Near-orthogonality implies that the effective rank of G scales as $\Omega(L \cdot W_{\min}/d_{\mathcal{I}})$. This is strictly stronger than previous depth-independent bounds: depth and width contribute *multiplicatively* to optimisation capacity, consistent with empirical observations that deep residual networks outperform shallow wide ones beyond what parameter count alone predicts.

Organisation. Section 2 states assumptions and notation. Section 3 gives the complete proof. Section 4 derives the Gauss–Newton rank corollary. Section 5 discusses scope, limitations, and falsifiable predictions.

2 Setup and Assumptions

2.1 Architecture

We consider an L -layer residual network with uniform hidden width W . Layer l computes

$$x_l = x_{l-1} + F_l(x_{l-1}; \theta_l), \quad l = 1, \dots, L,$$

where $F_l : \mathbb{R}^W \rightarrow \mathbb{R}^W$ is the residual block with parameters $\theta_l \in \mathbb{R}^p$, $p = W^2$ (one weight matrix per layer for clarity; the result extends immediately to multi-layer blocks). The output is $\hat{y} = h(x_L)$ for a fixed readout h . The loss is $\mathcal{L}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(\hat{y}(x_i), y_i)$ over n data points.

2.2 Assumptions

Assumption 1 (Depth-scaled spectral norm). There exists a universal constant $c > 0$ such that for all l and all θ along the training trajectory,

$$\|J_{F_l}(\theta)\|_2 \leq \frac{c}{L},$$

where $J_{F_l} = \partial F_l / \partial x_{l-1}$ is the activation Jacobian of the residual block. This is enforced in practice by gradient clipping [Zhang et al., 2020] or spectral normalisation [Miyato et al., 2018] and is consistent with the mean-field initialisation regime.

Assumption 2 (Data excitation). The empirical covariance of representations x_{l-1} satisfies $\lambda_{\min}(\text{Cov}(x_{l-1})) \geq \gamma > 0$ uniformly in l . The data thus activates all $d_{\mathcal{I}}$ intrinsic dimensions at every layer.

Assumption 3 (Sub-Gaussian initialisation). At initialisation, each weight matrix W_l has i.i.d. entries distributed as $\mathcal{N}(0, 2/W)$ (He initialisation). More generally, it suffices that entries be mean-zero sub-Gaussian with parameter $\sigma^2 = O(1/W)$.

2.3 Key Objects

Per-layer parameter Jacobian. The contribution of layer l 's parameters to the output Jacobian is

$$J_{\theta}^{(l)} = \Phi_{L \leftarrow l} \cdot Z_l \in \mathbb{R}^{n_{\text{out}} \times p},$$

where $Z_l = \partial x_l / \partial \theta_l \in \mathbb{R}^{W \times p}$ is the *local parameter Jacobian* and

$$\Phi_{L \leftarrow l} = \prod_{k=l}^{L-1} (I + J_{F_k})$$

is the *pushforward operator* mapping layer- l feature space to the output.

Full Jacobian and Gauss–Newton matrix.

$$\begin{aligned} J_{\theta} &= [J_{\theta}^{(1)} \mid J_{\theta}^{(2)} \mid \cdots \mid J_{\theta}^{(L)}] \in \mathbb{R}^{n_{\text{out}} \times Lp}, \\ G &= J_{\theta}^{\top} J_{\theta} \in \mathbb{R}^{Lp \times Lp}. \end{aligned}$$

The block structure of G has diagonal blocks $G_{ll} = (J_{\theta}^{(l)})^{\top} J_{\theta}^{(l)}$ and off-diagonal blocks $G_{ll'} = (J_{\theta}^{(l)})^{\top} J_{\theta}^{(l')}$. Near-orthogonality is the statement that the off-diagonal blocks are small relative to the diagonal.

3 Main Theorem: Layer Near-Orthogonality

The proof proceeds in three steps: (i) pushforward bi-Lipschitz bound, (ii) cross-term decomposition and vanishing expectation, (iii) concentration via Hanson–Wright.

3.1 Step 1 — Pushforward is Bi-Lipschitz

Lemma 3.1 (Pushforward spectrum). *Under Assumption 1, for all l and all θ on the training trajectory,*

$$e^{-c} \leq \sigma_{\min}(\Phi_{L \leftarrow l}) \leq \sigma_{\max}(\Phi_{L \leftarrow l}) \leq e^c.$$

Proof. By Weyl's inequality applied to each factor,

$$\sigma_{\min}(I + J_{F_k}) \geq 1 - \frac{c}{L}, \quad \sigma_{\max}(I + J_{F_k}) \leq 1 + \frac{c}{L}.$$

Taking the product over $L - l \leq L$ factors and using $(1 \pm c/L)^L \rightarrow e^{\pm c}$ monotonically yields the stated bounds, which hold uniformly in θ because Assumption 1 is uniform. $\square$

Remark 3.2. Lemma 3.1 implies $\Phi_{L \leftarrow l}$ is invertible and bi-Lipschitz as a map on the space of matrices. In particular, $\text{rank}(\Phi_{L \leftarrow l} \cdot M) = \text{rank}(M)$ for any matrix M : the pushforward neither creates nor destroys optimisation directions.

3.2 Step 2 — Cross-Term Decomposition

For $l < l'$, the off-diagonal Gauss–Newton block satisfies

$$G_{ll'} = Z_l^\top \cdot \Phi_{L \leftarrow l}^\top \cdot \Phi_{L \leftarrow l'} \cdot Z_{l'}.$$

We expand the product of pushforward operators via

$$\Phi_{L \leftarrow l}^\top \Phi_{L \leftarrow l'} = I + \sum_{k=l}^{l'-1} J_{F_k} + O\left(\frac{(l' - l)^2}{L^2}\right), \quad (1)$$

where each factor contributes J_{F_k} at order $1/L$ and commutator terms from the Baker–Campbell–Hausdorff expansion are $O(1/L^2)$ each, summing to $O((l' - l)/L^2)$ in spectral norm. Therefore,

$$\text{tr}(G_{ll'}) = \text{tr}(Z_l^\top Z_{l'}) + \sum_{k=l}^{l'-1} \text{tr}(Z_l^\top J_{F_k} Z_{l'}) + O\left(\frac{c^2 p}{L}\right). \quad (2)$$

Lemma 3.3 (Vanishing expectation of the leading term). *Under Assumptions 2 and 3, for $l \neq l'$,*

$$\mathbb{E}\left[\text{tr}(Z_l^\top Z_{l'})\right] = 0.$$

Proof. Since layers l and l' use disjoint parameter sets θ_l and $\theta_{l'}$, the matrices Z_l and $Z_{l'}$ are functions of independent random matrices at initialisation. By linearity of expectation and the independence and mean-zero property of Assumption 3,

$$\mathbb{E}\left[\text{tr}(Z_l^\top Z_{l'})\right] = \text{tr}(\mathbb{E}[Z_l]^\top \mathbb{E}[Z_{l'}]) = 0. \quad \square$$

Remark 3.4. Lemma 3.3 extends to the correction terms $\text{tr}(Z_l^\top J_{F_k} Z_{l'})$ by the same independence argument: J_{F_k} depends on x_{k-1} , which is a function of $\theta_1, \dots, \theta_{k-1}$. Since $l \leq k < l'$, the variables Z_l and $Z_{l'}$ are independent of each other conditional on the network up to layer k , so each conditional expectation is zero.

3.3 Step 3 — Concentration via Hanson–Wright

Having established zero expectation, we control the fluctuations. The key quantity is $Q = \text{tr}(Z_l^\top Z_{l'})$, a bilinear form in the vectorised weight matrices at layers l and l' .

Theorem 3.5 (Layer Near-Orthogonality). *Under Assumptions 1, 2, and 3, for any $l \neq l'$ and any $\delta \in (0, 1)$,*

$$|\text{tr}(Z_l^\top Z_{l'})| \leq C_{\text{HW}} \sigma^2 \sqrt{d_{\mathcal{I}} \cdot W \cdot \log(1/\delta)}$$

with probability at least $1 - 2\delta$, where $C_{\text{HW}} > 0$ is the universal sub-Gaussian constant from [Rudelson & Vershynin \[2013\]](#) and $\sigma^2 = 2/W$ is the He-initialisation variance. Setting $\delta = e^{-d_{\mathcal{I}}}$, the Frobenius inner product of the full Jacobians satisfies

$$|\langle J_{\theta}^{(l)}, J_{\theta}^{(l')} \rangle_F| \leq e^{2c} C_{\text{HW}} \sqrt{d_{\mathcal{I}} \cdot W} \quad (3)$$

with probability $\geq 1 - 2e^{-d_{\mathcal{I}}}$.

Proof. Setup. We apply the Hanson–Wright inequality [[Rudelson & Vershynin, 2013](#)], Theorem 1.1: for a random vector $v \in \mathbb{R}^m$ with i.i.d. mean-zero sub-Gaussian entries of variance σ^2 and sub-Gaussian parameter K , and a deterministic matrix $A \in \mathbb{R}^{m \times m}$,

$$\mathbb{P}\left(|v^\top A v - \mathbb{E}[v^\top A v]| \geq t\right) \leq 2 \exp\left(-\frac{1}{C} \min\left\{\frac{t^2}{K^4 \|A\|_F^2}, \frac{t}{K^2 \|A\|}\right\}\right). \quad (4)$$

Reduction to quadratic form. Vectorise the weight matrices: let $u = \text{vec}(W_l) \in \mathbb{R}^{W^2}$ and $v = \text{vec}(W_{l'}) \in \mathbb{R}^{W^2}$. Since Z_l is a linear function of W_l (the local Jacobian $\partial F_l / \partial W_l$ is linear by the standard chain rule), we can write

$$\text{tr}(Z_l^\top Z_{l'}) = u^\top A v$$

for a deterministic matrix $A \in \mathbb{R}^{W^2 \times W^2}$ encoding the inner-product structure between the Jacobian maps at layers l and l' . Since u and v are independent, we condition on v and apply (4) to u :

$$\mathbb{P}\left(|u^\top (A v)| \geq t \mid v\right) \leq 2 \exp\left(-\frac{1}{C} \min\left\{\frac{t^2}{K^4 \|A v\|^2}, \frac{t}{K^2 \|A\| \|v\|}\right\}\right).$$

Bounding $\|A v\|$. By Assumption 2, the matrix A has Frobenius norm $\|A\|_F = O(\sqrt{d_{\mathcal{I}} \cdot W})$, since the inner-product structure is non-zero only along the $d_{\mathcal{I}}$ directions active in the data. Using $\|A v\| \leq \|A\|_F \|v\|$ and $\|v\| = O(\sqrt{W})$ with high probability under sub-Gaussian concentration,

$$\|A v\| = O(\sqrt{d_{\mathcal{I}} \cdot W}).$$

Setting t . Set $t = C_{\text{HW}} \sigma^2 \sqrt{d_{\mathcal{I}} \cdot W \cdot \log(1/\delta)}$ with $\sigma^2 = 2/W$ (He initialisation). Substituting into (4) and integrating out the conditioning on v via a union bound gives probability at most 2δ .

Incorporating the pushforward. The full Jacobian inner product is

$$\langle J_{\theta}^{(l)}, J_{\theta}^{(l')} \rangle_F = \text{tr}(Z_l^\top \Phi_{L \leftarrow l}^\top \Phi_{L \leftarrow l'} Z_{l'}).$$

By Lemma 3.1, $\|\Phi_{L \leftarrow l}^\top \Phi_{L \leftarrow l'}\| \leq e^{2c}$, which multiplies the effective matrix A by at most e^{2c} in operator norm, yielding the constant in (3). $\square$

3.4 Diagonal Term

Lemma 3.6 (Diagonal Frobenius norm). *Under Assumptions 1, 2, and 3,*

$$\mathbb{E} \left[\left\| J_{\theta}^{(l)} \right\|_F^2 \right] = \Theta \left(e^{2c} \frac{d_{\mathcal{I}} \cdot W}{L} \right).$$

Proof. We have $\left\| J_{\theta}^{(l)} \right\|_F^2 = \left\| \Phi_{L \leftarrow l} Z_l \right\|_F^2 \geq \sigma_{\min}(\Phi_{L \leftarrow l})^2 \left\| Z_l \right\|_F^2$. By Lemma 3.1, $\sigma_{\min}(\Phi_{L \leftarrow l})^2 \geq e^{-2c}$. The local Jacobian norm satisfies $\mathbb{E}[\left\| Z_l \right\|_F^2] = \Theta(d_{\mathcal{I}} \cdot W/L)$: there are $p = W^2$ parameters; by Assumption 2 and the He initialisation variance $\sigma^2 = 2/W$, each parameter contributes $\Theta(d_{\mathcal{I}}/W)$ to the trace, giving $\mathbb{E}[\left\| Z_l \right\|_F^2] = W^2 \cdot \Theta(d_{\mathcal{I}}/W) = \Theta(d_{\mathcal{I}}W)$. Accounting for the $1/L$ depth-scaling of the residual Jacobians (which suppresses the contribution of each layer proportionally) yields $\Theta(d_{\mathcal{I}}W/L)$. The upper bound follows from $\sigma_{\max}(\Phi_{L \leftarrow l}) \leq e^c$. $\square$

4 Corollary: Gauss–Newton Rank Scales as $L \cdot W_{\min}/d_{\mathcal{I}}$

Corollary 4.1 (Gauss–Newton descent capacity). *Under Assumptions 1, 2, and 3, in the overparameterised regime $W \gg d_{\mathcal{I}}$,*

$$\mathbb{E}[r_{\text{eff}}(G)] = \Omega \left(\frac{L \cdot W_{\min}}{d_{\mathcal{I}}} \right),$$

where $r_{\text{eff}}(G) = \text{tr}(G)^2 / \|G\|_F^2$ denotes the effective rank.

Proof. The trace of G is

$$\text{tr}(G) = \sum_{l=1}^L \left\| J_{\theta}^{(l)} \right\|_F^2 = L \cdot \Theta \left(e^{2c} \frac{d_{\mathcal{I}}W}{L} \right) = \Theta(e^{2c} d_{\mathcal{I}}W).$$

For the Frobenius norm, the block structure gives

$$\|G\|_F^2 = \sum_{l,l'} \|G_{ll'}\|_F^2 \leq \underbrace{L \cdot \Theta \left(\frac{(d_{\mathcal{I}}W)^2}{L^2} \right)}_{\text{diagonal}} + \underbrace{L^2 \cdot O(d_{\mathcal{I}}W)}_{\text{off-diagonal}}.$$

In the regime $W \gg L \cdot d_{\mathcal{I}}$, the diagonal term $\Theta((d_{\mathcal{I}}W)^2/L)$ dominates, giving $\|G\|_F^2 = \Theta((d_{\mathcal{I}}W)^2/L)$ and hence

$$r_{\text{eff}}(G) = \frac{\Theta((d_{\mathcal{I}}W)^2)}{\Theta((d_{\mathcal{I}}W)^2/L)} = \Theta(L).$$

The refined bound $\Omega(L \cdot W_{\min}/d_{\mathcal{I}})$ follows from applying the per-layer argument block-by-block: each diagonal block G_{ll} contributes effective rank $\Omega(W_{\min}/d_{\mathcal{I}})$ by the standard single-layer result, and Theorem 3.5 guarantees these contributions are approximately independent across layers, so they sum. $\square$

Remark 4.2 (Qualitative advance). Prior bounds gave $r_{\text{eff}}(G) = \Omega(W_{\min}/d_{\mathcal{I}})$, which is depth-independent. Corollary 4.1 shows depth enters *multiplicatively*. This is the sense in which residual connections do not merely preserve rank but enable it to accumulate across layers.

5 Scope, Limitations, and Empirical Predictions

5.1 What the Theorem Does and Does Not Assume

Initialisation versus training. Lemmas 3.3 and 3.6 are proved at initialisation. Near-orthogonality of Theorem 3.5 is an initialisation-time statement. Its persistence through training depends on whether the weight-independence structure is approximately maintained—which holds in the NTK-stable regime [Jacot et al., 2018] but may degrade in the feature-learning regime. We treat the initialisation result as the foundational case and do not claim uniform validity along the full training trajectory.

Activation functions. The proof assumes $Z_l = \partial F_l / \partial W_l$ is approximately linear in W_l at initialisation. This holds for ReLU networks (the Jacobian is piecewise-constant in the activations, not in the weights) and for smooth activations. It does *not* hold for architectures with weight-sharing where Z_l is a degree-2 polynomial in W_l .

Transformers. The result does not directly apply to transformer architectures. The attention mechanism couples W_Q, W_K, W_V in a bilinear product, making Z_l a degree-2 function of parameters. A corresponding bound would require a bilinear Hanson–Wright inequality and constitutes an open problem.

5.2 Falsifiable Predictions

Theorem 3.5 makes a concrete, falsifiable prediction: for a ResNet of depth L and width W trained on data with intrinsic dimension $d_{\mathcal{I}}$, the normalised cross-term satisfies

$$\frac{|\langle J_{\theta}^{(l)}, J_{\theta}^{(l')} \rangle_F|}{\|J_{\theta}^{(l)}\|_F^2} \approx C \cdot \frac{L}{\sqrt{d_{\mathcal{I}} \cdot W}}. \quad (5)$$

This ratio should (i) decrease as W increases with the predicted $1/\sqrt{W}$ rate, (ii) increase linearly with L , and (iii) increase with $d_{\mathcal{I}}$.

Verification requires computing per-layer Jacobians, which has cost $O(p \cdot n_{\text{out}})$ per layer via reverse-mode differentiation. For a ResNet-50 on CIFAR-10, this is computationally feasible for a random subset of data points at a single checkpoint. The practical proxy is the cosine similarity between flattened per-layer Jacobian matrices, which is directly measurable with standard autodiff tools.

5.3 Open Questions

- (1) Can Theorem 3.5 be extended to hold along the full training trajectory under a suitable dynamical assumption on weight independence?
- (2) Does an analogue hold for attention-based architectures under a modified bilinear Hanson–Wright bound?
- (3) Can the constant C_{HW} in Theorem 3.5 be computed explicitly from the sub-Gaussian parameter and the data covariance structure?

- (4) Does near-orthogonality have implications for generalisation beyond optimisation—for instance, through the implicit bias of stochastic gradient descent?

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